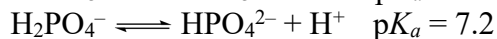
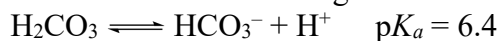


Chapter 17 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Consider a solution consisting of the following two buffer systems:



At pH 6.8, which one of the following is true of the relative amounts of acid and conjugate base present?

- A) $[\text{H}_2\text{CO}_3] > [\text{HCO}_3^-]$ and $[\text{H}_2\text{PO}_4^-] > [\text{HPO}_4^{2-}]$
- B) $[\text{HCO}_3^-] > [\text{H}_2\text{CO}_3]$ and $[\text{H}_2\text{PO}_4^-] > [\text{HPO}_4^{2-}]$
- C) $[\text{H}_2\text{CO}_3] > [\text{HCO}_3^-]$ and $[\text{HPO}_4^{2-}] > [\text{H}_2\text{PO}_4^-]$
- D) $[\text{HCO}_3^-] > [\text{H}_2\text{CO}_3]$ and $[\text{HPO}_4^{2-}] > [\text{H}_2\text{PO}_4^-]$
- E) $[\text{H}_2\text{CO}_3] = [\text{HCO}_3^-]$ and $[\text{H}_2\text{PO}_4^-] > [\text{HPO}_4^{2-}]$

2. How much solid NaCN must be added to 1.0 L of a 0.5 M HCN solution to produce a solution with pH 7.0? $K_a = 6.2 \times 10^{-10}$ for HCN.

- A) 0.0031 g
- B) 160 g
- C) 0.15 g
- D) 24 g
- E) 11 g

3. When 200. mL of 2.0 M NaOH(aq) is added to 500. mL of 1.0 M HCl(aq), the pH of the resulting mixture is closest to _____.

- A) 1.0
- B) 3.0
- C) 7.0
- D) 8.0
- E) 13.0

4. AgBr(s) would have the lowest solubility in _____.

- A) pure H₂O
- B) 0.040 M NH₃
- C) 0.040 M KBr
- D) 0.040 M SrBr₂
- E) 0.010 M AgNO₃

5. A 200.0-mL sample of the weak acid H₃A (0.100 M) is titrated with 0.100 M NaOH. After 200.0 mL NaOH is added, what is the major species in the titration? (Water is always assumed to be a major species.)

- A) HA²⁻, A³⁻
- B) HA²⁻
- C) A³⁻
- D) H₂A⁻, HA²⁻
- E) H₂A⁻

6. Silver acetate (AgC₂H₃O₂) is a sparingly soluble salt with $K_{sp} = 1.9 \times 10^{-3}$. Consider a saturated solution in equilibrium with the solid salt. Compare the effects on the solubility of adding to the solution either the acid HNO₃ or the base NH₃.

- A) NH₃ would increase the solubility, but HNO₃ would decrease it.
- B) NH₃ would decrease the solubility, but HNO₃ would increase it.
- C) NH₃ would increase the solubility, but HNO₃ would have virtually no effect.
- D) Either substance would decrease the solubility.
- E) Either substance would increase the solubility.

7. Which of the following compounds has the lowest solubility, in moles per liter, in water?

- A) MgC₂O₄ $K_{sp} = 8.6 \times 10^{-5}$
- B) PbSO₄ $K_{sp} = 1.3 \times 10^{-8}$
- C) CdS $K_{sp} = 1.0 \times 10^{-28}$
- D) Al(OH)₃ $K_{sp} = 2 \times 10^{-32}$
- E) Sn(OH)₂ $K_{sp} = 3 \times 10^{-27}$

8. What is the pH of a solution that results when 0.012 mol HNO₃ is added to 610.0 mL of a solution that is 0.26 M in aqueous ammonia and 0.46M in ammonium nitrate. Assume no volume change. (K_b for NH₃ = 1.8×10^{-5} .)

- A) 8.96
- B) 9.56
- C) 5.04
- D) 9.01
- E) 4.44

Chapter 17 Homework

9. Calculate the pH at the equivalence point for the titration of 100.0 mL of 0.10 *M* HCOOH, by 0.10 *M* NaOH. (K_a for HCOOH = 1.8×10^{-4})

- A) 7.00 B) 10.25 C) 2.09 D) 5.38 E) 8.22

10. You are given 5.00 mL of an H₂SO₄ solution of unknown concentration. You divide the 5.00-mL sample into five 1.00-mL samples and titrate each separately with 0.1000 *M* NaOH. In each titration, the H₂SO₄ is completely neutralized. The average volume of NaOH solution used to reach the endpoint is 15.3 mL. What was the concentration of H₂SO₄ in the 5.00-mL sample?

- A) 0.306 *M* B) 0.765 *M* C) 3.06 *M* D) 1.53×10^{-3} *M* E) 1.53 *M*

	K_a
HC ₃ H ₅ O ₃ (aq)	8.3×10^{-4}
CH ₃ NH ₃ ⁺ (aq)	2.3×10^{-11}

11. The acid-dissociation constants of HC₃H₅O₃(aq) and CH₃NH₃⁺(aq) are given in the table above. Which of the following mixtures is a buffer with a pH of approximately 3 ?

- A) A mixture of 100. mL of 0.1 *M* CH₃NH₃⁺ and 50. mL of 0.1 *M* NaOH
B) A mixture of 100. mL of 0.1 *M* HC₃H₅O₃ and 50. mL of 0.1 *M* NaOH
C) A mixture of 100. mL of 0.1 *M* NaC₃H₅O₃ and 100. mL of 0.1 *M* NaOH
D) A mixture of 100. mL of 0.1 *M* CH₃NH₃Cl and 100. mL of 0.1 *M* CH₃NH₂
E) A mixture of 100. mL of 0.1 *M* CH₃NH₃Cl and 100. mL of 0.1 *M* HCl

12. You have a solution of 0.10 *M* Cl⁻ and 0.10 *M* CrO₄²⁻. You add 0.10 *M* silver nitrate dropwise into the solution. K_{sp} for Ag₂CrO₄ is 9.0×10^{-12} and for AgCl is 1.6×10^{-10} . Which of the following will precipitate first?

- A) silver nitrate
B) silver chromate
C) cannot be determined from the information given
D) silver chloride
E) none of these

13. Equal volumes of 0.1 *M* HCl and 0.1 *M* HF are titrated with 0.1 *M* NaOH, respectively. Which of the following would be equal for both titrations?

- A) the volume of NaOH added to reach the equivalence point
B) the pH at the equivalence point
C) the initial pH
D) the pH at the halfway point
E) two of the above

14. Which solution has the greatest buffering capacity?

- A) 0.335 *M* NH₃ and 0.100 *M* NH₄Cl
B) 0.085 *M* NH₃ and 0.090 *M* NH₄Cl
C) 0.600 *M* NH₃ and 0.600 *M* NH₄Cl
D) 0.200 *M* NH₃ and 0.565 *M* NH₄Cl
E) They are all buffer solutions and would all have the same capacity.

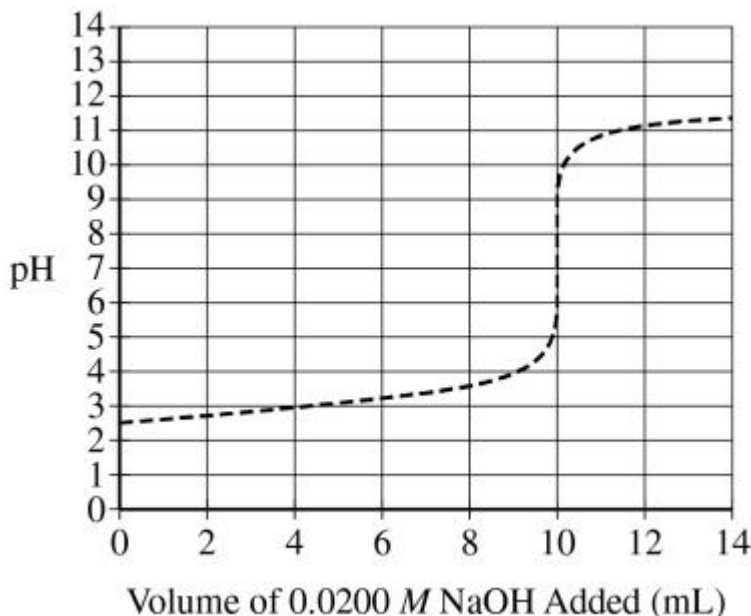
15. For the titration of a weak base with a strong acid, which of the following indicators would be best to visualize the equivalence point?

- A) Cresol red ($pK_a = 1.0$) B) Methyl red ($pK_a = 5.0$)
C) Thymol blue ($pK_a = 8.9$) D) Phenolphthalein ($pK_a = 9.4$)
E) Alizarin yellow ($pK_a = 11.2$)

Chapter 17 Homework

Part B Short questions (15 points)

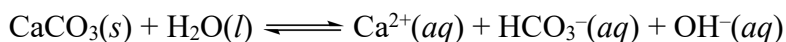
16. (7 points) A student titrates 20.0 mL of 0.0100 *M* salicylic acid ($\text{HC}_7\text{H}_5\text{O}_3$) with 0.0200 *M* NaOH, using a probe to monitor the pH of the solution. The data are plotted producing the following titration curve.



- (a) Using the information in the graph, estimate the $\text{p}K_a$ of salicylic acid ($\text{HC}_7\text{H}_5\text{O}_3$).
(b) When the pH of the titration mixture is 4.00, is there a higher concentration of the weak acid, $\text{HC}_7\text{H}_5\text{O}_3$, or its conjugate base, $\text{C}_7\text{H}_5\text{O}_3^-$, in the flask? Justify your answer.
(c) The student performs a second titration, this time titrating 20.0 mL of a 0.0100 *M* benzoic acid ($\text{HC}_7\text{H}_5\text{O}_2$) solution with 0.0200 *M* NaOH. Sketch the curve that would result from this titration of benzoic acid on the graph of the answer sheet, which already shows the original curve from the titration of 20.0 mL of 0.0100 *M* salicylic acid ($\text{HC}_7\text{H}_5\text{O}_3$). The K_a for benzoic acid is 6.3×10^{-5} .

17. (8 points) The solubility of CaCO_3 is pH dependent.

- (a) Calculate the molar solubility of CaCO_3 ($K_{sp} = 4.5 \times 10^{-9}$) neglecting the acid–base character of the carbonate ion.
(b) Use the K_b expression for the CO_3^{2-} ion (K_{a2} for H_2CO_3 is 5.6×10^{-11}) to calculate the equilibrium constant for the reaction



- (c) If we assume that the only sources of Ca^{2+} , CO_3^{2-} , and OH^- ions are from the dissolution of CaCO_3 , what is the molar solubility of CaCO_3 using the preceding expression? What is the pH?
(d) If the pH is buffered at 7.5, what is the molar solubility of CaCO_3 ?

Chapter 17 Homework Answer Sheet

Name:

Student ID:

Instructor:

Score:

Part A Multiple choice (15 points)

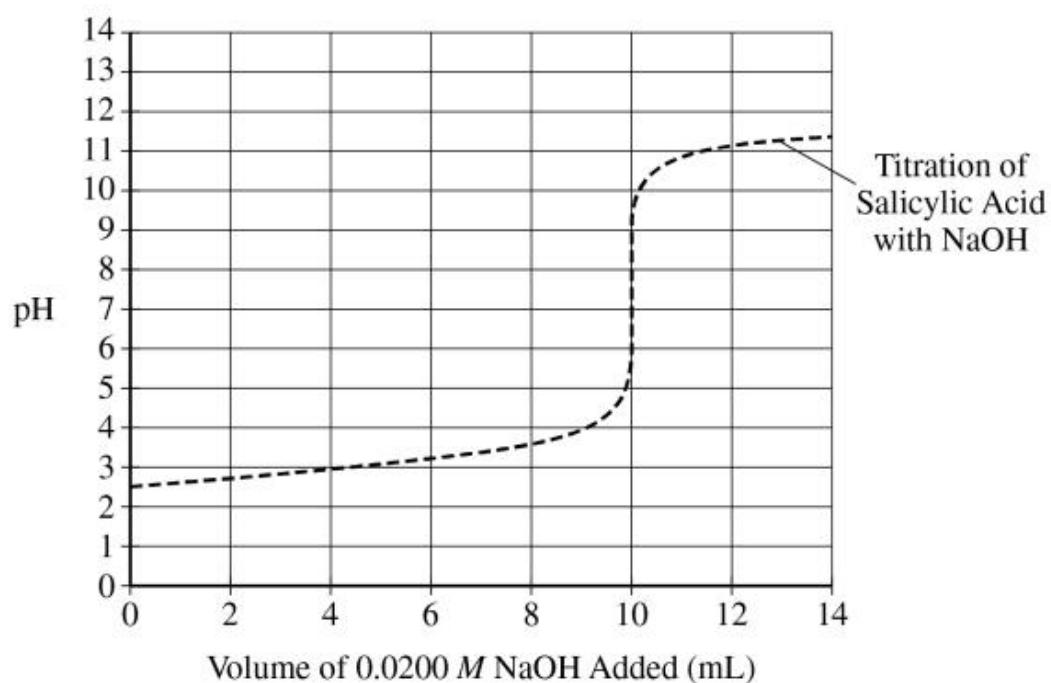
1-5 _____

6-10 _____

11-15 _____

Part B Short questions (15 points)

16.



17.